



**Verified Carbon
Standard**

LAMANTIN

COMMUNITY-BASED MANGROVE
CONSERVATION AND RESTORATION IN
GUINEA



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1. PROJECT DETAILS

1.1. Summary Description of the Project

Lamantin mangrove restoration and conservation project, developed by Blue Coast Carbon, is a community-based large-scale mangrove conservation project in Guinea, West Africa.

The project will be located in an area along the northern coasts of Guinea, covering a total of 73,751 hectares of mangrove forests.

Mangroves are coastal forests that grow where ocean water, freshwater, and land meet. They are amongst the planet's most productive and complex ecosystems, and are recognized for their superior carbon sequestration capabilities compared to terrestrial forests. Throughout West Africa, the livelihoods of coastal populations depend heavily on access to natural resources. Mangroves are integral to many of those resources, providing wood and non-wood forest products, coastal protection, conservation of biological diversity, provision of habitat, spawning grounds and nutrients for a variety of fish and shellfish, and salt production. West Africa's coastlines have some of the highest and most rapidly growing populations. Many communities rely on mangrove wood as a primary fuel source for curing fish and other purposes, and urban expansion and intensifying demands for charcoal, fuel wood, and land for agriculture are growing drivers of mangrove deforestation and degradation¹. These factors, combined with rising sea levels, erosion from extreme weather, and more intense storm surges, represent significant and growing threats to mangroves².

The overall project objective is to generate high-quality carbon credits based on greenhouse gases (GHGs) emission reduction and avoidance, that will sustainably finance the project and result in economic benefits for the local communities. GHGs emission reductions are expected to be generated by avoiding unplanned deforestation by providing alternative income to local communities through the implementation of a series of activities, including those supporting the use of efficient cookstoves, solar salt production systems, education campaigns, and supporting the development of apiculture, aquaculture and ecotourism.

Carbon dioxide (CO₂) removals will be generated by restoring areas of degraded mangrove forests with the involvement of local communities, supporting local livelihoods and alleviating poverty. An ecosystem-based approach will be used for habitat restoration, aiming at improving hydrology and environmental conditions, with minimal need for re-planting.

The project will restore a total of 2,725 hectares of degraded mangrove forests and protect a total of 73,751 hectares from deforestation, contributing to the protection of coastal communities from climate change, the conservation of the world's biodiversity and to mitigating climate change.

¹ USAID, 2014, Workshop Report: West Africa Regional Mangroves and Climate Change, U.S. Agency for International Development

² Corcoran, E., Ravilious, C., and Skuja, M., 2007, Mangroves of western and central Africa: UNEP/Earthprint 26

The project's climate benefits over the crediting period include the sequestration and avoided emission of 3,374,658 tCO₂e.

1.2. Audit History

The project does not have an audit history yet. The first audit will be completed at the time of project validation and initial verification.

1.3. Sectoral Scope and Project Type

Sectoral scope	14 Agriculture, forestry, and other land use (AFOLU)
AFOLU project category ³	Wetlands Restoration and Conservation (WRC)
Project activity type	Restoring wetland ecosystem (RWE) combined with ARR Conservation of intact wetlands (CIW) – Avoiding Unplanned Wetland Degradation – combined with REDD+

1.4. Project Eligibility

1.4.1. General eligibility

The project is eligible under the scope of the VCS Program, as it aims at enhancing the sequestration and avoiding the emission of carbon dioxide (CO₂), methane (CH₄) and nitrous oxide (N₂O) by mangrove forest conservation and ecosystem restoration.

Project activities are supported by the VCS methodologies VM0033 and VM0007. Furthermore, the project does not involve any of the activity listed in Table 1 of the VCS Standard v4.7.

The project will initiate the pipeline listing process within three years of the project start date and complete validation within eight years of the project start date.

The applied methodologies VM0033 and VM0007 are applicable to WRC and REDD+ projects under the VCS Program.

³ See Appendix 1 of the VCS Standard

1.4.2. AFOLU project eligibility

The project falls under the AFOLU project categories WRC, ARR, and REDD+ as defined in Appendix 1 Eligible AFOLU project categories of the VCS Standard v4.7. Table 1, Table 2 and Table 3 present the VCS Standard's requirements and how the project will comply with them.

Table 1 Compliance with eligibility conditions of WRC projects (from Appendix 1 of the VCS Standard V4.7).

Eligibility condition	Compliance
Eligible WRC activities are those that increase net GHG removals by restoring wetland ecosystems or that reduce GHG emissions by rewetting or avoiding the degradation of wetlands.	The project activities will increase net GHG removals by restoring degraded mangrove forests and will reduce GHG emissions by avoiding additional degradation and deforestation.
The project area shall meet an internationally accepted definition of wetland, such as from the IPCC, Ramsar Convention on Wetlands, those established by law or national policy, or those with broad agreement in the peer-reviewed scientific literature for specific countries or types of wetlands.	'Wetlands include any land that is covered or saturated by water for all or part of the year, and that does not fall into the Forest Land, Cropland, or Grassland categories'(IPCC)⁴. The project involves areas of mangrove forests in coastal wetlands, i.e. daily affected by astronomical tides and saline water.
Common wetland types include peatland, salt marsh, tidal freshwater marsh, mangroves, wet floodplain forests, prairie potholes and seagrass meadows.	The project area is mainly covered by mangrove forests.
Activities that generate net reductions of GHG emissions from wetlands are eligible as WRC projects or combined category projects (such as REDD on peatland).	The project will increase net GHG removals by restoring the mangrove ecosystem, combined with conservation of existing mangrove forests.

⁴ Chapter 7 Wetlands, Volume 4, 2006 IPCC Guidelines

Activities that actively lower the water table depth in wetlands are not eligible.	Project activities will not lower the water table depth.
Eligible WRC activities include <u>Restoring Wetland Ecosystems (RWE)</u> : This category includes activities that reduce GHG emissions or increase carbon sequestration in a degraded wetland through restoration conditions, sediment supply, salinity characteristics, water quality and/or native plant communities. For the purpose of these requirements, restoration activities are those that result in the reestablishment of ecological processes, functions, and biotic and/or abiotic linkages that lead to persistent, resilient systems integrated within the landscape.	The project will restore degraded wetlands by means of activities aimed at restoring natural conditions to allow for spontaneous habitat recovery, such as removal of pollution and restoration of natural hydrological conditions to allow for optimum mangrove growth. The abiotic conditions will be carefully assessed before the start of the activities to identify the most suitable interventions.
<u>Conservation of Intact Wetlands (CIW)</u> : This category includes activities that reduce GHG emissions by avoiding degradation and/or the conversion of wetlands that are intact or partially altered while still maintaining their natural functions, including hydrological conditions, sediment supply, salinity characteristics, water quality and/or native plant communities.	The project involves activities aimed at protecting intact wetlands thus reducing GHGs emissions by avoiding degradation of mangrove forests. These include the promotion of sustainable systems for salt production and efficient cook stoves, together with education and awareness programmes.
<u>Avoiding Planned Wetland Degradation (APWD)</u>: This activity reduces GHG emissions by avoiding degradation of wetlands, or further degradation in partially drained wetlands that are legally authorized and documented for conversion. <u>Avoiding Unplanned Wetland Degradation (AUWD)</u>: This activity reduces GHG emissions by avoiding unplanned degradation of wetlands, or by avoiding further degradation in partially degraded wetlands.	The activities of this project fall under the scope of Avoiding Unplanned Deforestation and/or Degradation, as there are no set plans for any activity that would cause wetland degradation.
Activities where drainage is continued or maintained are not eligible. This includes, for example, projects that require the maintenance of drainage channels to maintain the pre-project drainage level on a partially drained peatland (e.g.,	Drainage will not be performed.

where periodic deepening may be needed to counteract peat subsidence).	
Projects that allow selective harvesting that results in a lowering of the water table depth (e.g., by extracting timber using drainage canals) or affects the ability of vegetation to act as a major hydrological regulation device (e.g., extracting trees which support the peat body) are not eligible.	This project will not allow selective harvesting.
Activities that generate net GHG emission reductions by combining other AFOLU project activities with wetlands restoration or conservation activities are eligible as WRC combined projects. RWE may be implemented without further conversion of land use, or it may be combined with ARR, ALM, IFM, REDD or ACoGS activities, referred to as ARR+RWE, ALM+RWE, IFM+RWE, REDD+RWE or ACoGS+RWE, respectively. CIW may be implemented on non-forest land or combined with IFM, REDD or ACoGS activities, referred to as IFM+CIW, REDD+CIW or ACoGS+CIW, respectively.	The baseline scenario in this project is degraded wetland and forest with deforestation and degradation as the main land use. Project activities include restoration of wetlands (mangroves) and revegetation. There is no further conversion of land use. Therefore, this project is a WRC project combining ARR and RWE.
Combined category projects shall use the relevant WRC requirements and the respective AFOLU project category requirements for quantifying GHG emissions/removals unless the former may be deemed <i>de minimis</i> or conservatively excluded.	See section 3

Table 2 Compliance with eligibility conditions of ARR projects (from Appendix 1 of the VCS Standard V4.7).

Eligibility condition	Compliance
Eligible ARR activities are those that increase carbon sequestration and/or reduce GHG emissions by establishing, increasing, or restoring vegetative cover (forest or non-forest) through the	The project will increase carbon sequestration and reduce GHG emissions by restoring mangrove forest cover through assisting natural regeneration of vegetation and planting where necessary. This will increase biomass production and soil carbon

planting, sowing, or human-assisted natural regeneration of woody vegetation.	sequestration, while conserving the established forests.
Eligible ARR projects may include timber harvesting in their management plan.	The project activities do not include timber harvesting.

Table 3 Compliance with eligibility conditions of REDD+ projects (from Appendix 1 of the VCS Standard V4.7).

Eligibility condition	Compliance
The project area shall meet an internationally accepted definition of forest, such as those based on UNFCCC host-country thresholds or FAO definitions and shall qualify as forest for a minimum of 10 years before the project start date. The definition of forest may include mature forests, secondary forests, and degraded forests. Under the VCS Program, secondary forests are considered to be forests that have been cleared and have recovered naturally and that are at least 10 years old and meet the lower bound of the forest threshold parameters at the start of the project.	According to the UNFCCC, a forest is defined as an area of land spanning more than 0.05-1.0 hectares with tree crown cover (or equivalent stocking level) of more than 10-30% with trees having the potential to reach a minimum height of 2-5 meters at maturity in situ. The mangrove forests in the project area have met such requirements over the last 10 years.
Activities covered under the REDD project category are those that are designed to stop planned (designated and sanctioned) deforestation or unplanned (unsanctioned) deforestation and/or degradation. Avoided planned degradation is classified as IFM.	Activities for this project are designed to stop unplanned deforestation and/or degradation.
Activities that stop unsanctioned deforestation and/or illegal degradation (such as removal of fuelwood or timber extracted by non-concessionaires) on lands that are legally sanctioned for timber production are eligible as REDD activities. However, activities that reduce or stop logging only, followed by protection, on forest lands legally designated or sanctioned for forestry	The activities do not stop logging and are not on the IFM scope. They stop unsanctioned deforestation and unplanned degradation.

activities are included within IFM. Projects that include both avoided unplanned deforestation and/or degradation as well as stopping sanctioned logging activities, shall follow the REDD guidelines for the unplanned deforestation and/or degradation and the IFM guidelines for the sanctioned logging activities, and shall follow the requirements set out in Section 3.6.2.	
Eligible REDD activities include: 1) Avoiding Planned Deforestation (APD): This category includes activities that reduce net GHG emissions by permanently stopping or reducing deforestation on forest lands that are legally authorized and documented for conversion. Planned deforestation encompasses activities where a forest system would have been cleared and replaced by a different forest system with a lower carbon stock and where the recovery of timber was not the primary objective of the initial forest clearance. 2) Avoiding Unplanned Deforestation and/or Degradation (AUDD): This category includes activities that reduce net GHG emissions by stopping deforestation and/or degradation of degraded to mature forests that would have occurred in any forest configuration.	The activities of this project fall under the scope of Avoiding Unplanned Deforestation and/or Degradation

1.4.3. Transfer project eligibility

N/A. The present project is not a transfer project

1.5. Project Design

Indicate if the project has been designed as:

- ☐ Single location or installation
- ☐ Multiple locations or project activity instances (but not a grouped project)

✓ Grouped project

1.5.1. Grouped project design

This is a grouped project with the following eligibility criteria for the inclusion of new project activity instances:

	Eligibility condition	Compliance
Baseline scenario and additionality	Grouped projects shall specify one or more clearly defined geographic areas within which project activity instances may be developed. Such geographic areas shall be specified using geodetic polygons as set out in Section 3.11 below. Geographic areas with no initial project activity instances shall not be included in the project unless it can be demonstrated that the same (or at least as conservative) baseline scenario and rationale for the demonstration of additionality is applicable to such an area as a geographic area that does include initial project activity instances.	The grouped project area as well as the first instance are detailed in the “Project Location” section below. The future instances will soon be defined but the baseline scenario is the same in the entire grouped project area as well as the additionality demonstration (see below).
	Determination of baseline scenario and demonstration of additionality are based upon the initial project activity instances. The initial project activity instances are those that are included in the project description at validation and shall include all project activity instances currently implemented on the issue date of the project description. The initial project activity instances may also include any instances of the project activity that have been planned and developed to a sufficient level of	At the present stage, only the first instance will be included in the PDD. The baseline scenario and demonstration of additionality are the same for the entire grouped project area as for the initial project activity instance.

	detail to enable their assessment at validation	
	As with non-grouped projects, grouped projects may incorporate multiple project activities. Where a grouped project includes multiple project activities, the project description shall designate which project activities may occur in each geographic area.	Project activities are detailed in the “Description of the Project Activities” section below, while their geographic location is reported in section 3.3.
	The baseline scenario for a project activity shall be determined for each designated geographic area, in accordance with the methodology applied to the project. Where a single baseline scenario cannot be determined for a project activity over the entirety of a geographic area, the geographic area shall be redefined or divided such that a single baseline scenario can be determined for the revised geographic area or areas.	The grouped project area is considered one project area, with one same baseline scenario for the entire grouped project.
	The additionality of the initial project activity instances shall be demonstrated for each designated geographic area, in accordance with the methodology applied to the project. Where the additionality of the initial project activity instances within a particular geographic area cannot be demonstrated for the entirety of that geographic area, the geographic area shall be redefined or divided such that the additionality of the instances occurring in the revised geographic area or areas can be demonstrated.	The grouped project area is considered one project area, with the same additionality demonstration for the entire grouped project.

	Where factors relevant to the determination of the baseline scenario or demonstration of additionality require assessment across a given area, the area shall be, at a minimum, the grouped project geographic area. Examples of such factors include, inter alia, common practice; laws, statutes, regulatory frameworks, or policies relevant to demonstration of regulatory surplus; determination of regional grid emission factors; and historical deforestation and degradation rates.	The grouped project area is considered one single project area, with one baseline scenario and one additionality demonstration for the entire grouped project. More details on the baseline scenario and the additionality demonstration are provided in the relevant sections.
New project activity instance	Grouped projects shall include one or more sets of eligibility criteria for the inclusion of new project activity instances. At least one set of eligibility criteria for the inclusion of new project activity instances shall be provided for each combination of project activity and geographic area specified in the project description. Where grouped projects include multiple baseline scenarios or demonstrations of additionality, such projects will require at least one set of eligibility criteria for each combination of baseline scenario and demonstration of additionality specified in the project description.	As the grouped project area is one single project area with the same additionality demonstration and baseline scenario, as well as the same activities, only one set of eligibility criteria is needed for all instances, and they are defined in the relevant sections above.
	New project activity instances shall: 1) Occur within one of the designated geographic areas specified in the project description. 2) Conform with at least one complete set of eligibility criteria for the inclusion of new project activity	1) Future project activity instances will be within the grouped project boundaries, as defined in the “Project location” section. 2) New instances will follow the set of eligibility criteria described above.

instances. Partial conformance with multiple sets of eligibility criteria is insufficient. 3) Be included in the monitoring report with sufficient technical, financial, geographic, and other relevant information to demonstrate conformance with the applicable set of eligibility criteria and enable evidence gathering by the validation/verification body. 4) Be included in an updated project description, with updated project location information, which shall be validated at the time of verification against the applicable set of eligibility criteria. 5) Have evidence of project ownership, in respect of each project activity instance, held by the project proponent from the respective start date of each project activity instance (i.e., the date upon which the project activity instance began reducing or removing GHG emissions). 6) Have a start date that is the same as or later than the grouped project start date. 7) Be eligible for crediting from the start date of the project activity instance through to the end of the project crediting period (only). 8) Only eligible for crediting from the start of the verification period in which they were added to the grouped project. 9) Not be or have been enrolled in another VCS project.	3) Future instances will be included in the monitoring report. 4) Future instances will be included in an updated project description document. 5) The project ownership will be the same for future instances as it is for the first instance. 6) The future instances will have a later start date than the start of the grouped project. 7) The future instances will only be eligible for crediting from the start date of the project activity instance through to the end of the project crediting period. 8) The future instances will only be eligible for crediting from the start of the verification period in which they were added to the grouped project. 9) Future instances will not/have not been enrolled in other VCS projects. 10) Yes.
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	10) Adhere to the clustering and capacity limit requirements for multiple project activity instances set out in 3.6.8 - 3.6.9.	
	Where inclusion of a new project activity instance necessitates the addition of a new project proponent to the project, such instances shall be included in the grouped project description within five years of the project activity instance start date, where the project activity is an AFOLU activity.	The project proponent do not foresee the addition of a new proponent to the project. However, if the inclusion of a new instance requires that, such instances shall be included in the grouped project description within five years of the project activity instance start date.

1.6. Project Proponent

Organization name	Blue Coast Carbon
Contact person	James Harrison
Title	President
Address	100 Avenue de L'Adour Centre Olatu, Anglet, 64600 France
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1.7. Other Entities Involved in the Project

Organization name	Ministère de l'Environnement et du Développement Durable
Role in the project	Partner
Contact person	Dr Karim SAMOURA
Title	Secretary General
Address	Coléya Lanséboungny, BP 761 Guinea
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Email	sg.medd@gouvernement.gov.gn

1.8. Ownership

To manage the project and legally secure ownership over the carbon credits generated by the project, Blue Coast Carbon is seeking a concession agreement with the government. The Ministry of Environment and Sustainable Development (MEDD) issued Blue Coast Carbon a letter on the 16th December 2025 approving the development and finalisation of this project. The letter gives Blue Coast Carbon the right to list the project as under development.

Mangrove forests in Guinea are classified as public property under the public maritime domain. This means they are owned by the State and cannot be privately owned or transferred. The primary challenges for obtaining ownership of mangroves include:

- Inalienability: the public maritime domain is inalienable, imprescriptible, and unseizable, meaning the state cannot cede ownership or allow private acquisition through occupation.
- Legal restrictions: existing laws, such as the Environmental Code and Maritime Code, explicitly state that mangroves are part of the public domain, preventing any form of private ownership or long-term leases.

According to the Environmental Code, the State of Guinea can conduct conventions with third parties that can contribute to the protection of the environment. Entities can obtain temporary occupation permits or

concessions from the state, allowing them to manage or exploit mangrove areas without transferring ownership.

This section will be appropriately updated in the PDD submitted for validation.

1.9. Project Start Date

Project start date	1 st April 2026
Justification	The project start date marks the launch of project activities after the completion of the first step of the FPIC process in the first project instance, which is expected to be completed before the end of March 2026.

1.10. Project Crediting Period

Crediting period	☐ Seven years, twice renewable ☐ Ten years, fixed ☒ Other:40 years, renewable up to 100 years This is compliant with VCS Program (v4.7) requirements for AFOLU projects, i.e. minimum 20 years and maximum 100 years
Start and end date of first or fixed crediting period	To be completed in the PDD version submitted for validation

1.11. Project Scale and Estimated GHG Emission Reductions or Removals

- ✓ < 300,000 tCO₂e/year (project)
- ≥ 300,000 tCO₂e/year (large project)

1.12. Description of the Project Activity

The overall project objective is to generate high-quality carbon credits based on greenhouse gases (GHGs) emission reduction and removal, that will sustainably finance the project and result in economic benefits

for the local communities. GHGs emission reductions are expected to be generated by avoiding unplanned deforestation by providing alternative income to local communities through the implementation of a series of activities, including ecotourism, beekeeping, aquaculture, and education campaigns. Alternatives to the current cooking methods and salt production systems will be also introduced to reduce pressure on mangrove forests.

Carbon dioxide (CO₂) removals will be generated by restoring areas of degraded mangrove forests and recovering mangrove vegetation cover with the involvement of local communities, supporting local livelihoods and alleviating poverty.

The project will protect a total of 73,751 hectares from deforestation, contributing to the protection of coastal communities from climate change, the conservation of floral and faunal diversity and climate change mitigation.

The project's climate benefits over the crediting period include the net avoided emissions of 3,374,658 tCO₂e.

Blue Coast Carbon will deploy large scale reforestation with community participation through the following strategic blue carbon conservation activities:

1. Mangrove conservation awareness and education, and training of communities in sustainable forest management;
2. Mangrove restoration and sustainable mangrove forest management with community-led monitoring and surveillance;
3. Promoting the adoption of fuel-efficient small wood cookstoves and sustainable solutions for fish smoking and salt production;
4. Promoting the adoption of sustainable alternative livelihoods, such as ecotourism, beekeeping and aquaculture, for income diversification.

Blue Coast Carbon will conduct research, engage with community partners, and pilot different project activities during the early stages of project implementation (i.e., in 2026). The most successful project activities will be piloted through continuous community consultation. Community engagement will be carried out following the guidelines established by FPIC. A community consultation plan and protocols to ensure FPIC of indigenous peoples and local communities (IPLCs) living in and around the project area will be developed with the collaboration of local partners.

The identified income diversification activities will be piloted in collaboration with experts following thorough research to determine best practices and learn from the experiences of other projects around the world.

Blue Coast Carbon has been actively working on the feasibility study in Guinea since September 2024. The project proponent has been active on the ground in Guinea since March 2025. Project activities in Guinea started with stakeholder mapping, partner selection and the appointment of representatives in Conakry, Guinea.

The project activities undertaken so far include:

- the full assessment of the Guinea coastline using Blue Coast Carbon's proprietary data pipeline and ESA and NASA Earth Observation imagery;
- assessment of the carbon sequestration potential of the restoration of degraded mangrove forest areas and estimation of the emissions potentially avoided by the conservation of pristine mangrove forests within the proposed project areas. The quantification of carbon stocks of the proposed project areas was performed by estimating above- and belowground mangrove biomass and soil organic carbon;
- Specific project area analyses and full country analysis necessary to confirm the technical and financial feasibility of the project;
- Briefing of government stakeholders on our company and project approach;
- Financial file and evidence submitted to the Ministry of Finance;
- A Memorandum of Understanding for the development of blue carbon projects in the project area was signed by the Ministry of The Environment and Sustainable Development (MEDD).

Activity plan

The proposed project activities include the following components:

1. Site preparation:

This step involves carrying out all survey and data collection activities necessary to complete the baseline assessment. In particular:

- ground survey and mapping of mangrove extension and degradation areas
- in-situ validation of remotely-sensed data
- forest inventory, e.g. map distribution of species in the project area, identification of endangered species and baseline biodiversity assessment, mangrove propagules supply assessment
- identification of activities needed for ecosystem restoration, e.g. hydrological intervention, pollution removal. Planting will be carried out only if needed, as the priority will be given to habitat restoration, because this is known to have higher success rates and faster recovery times.

Seven species of mangrove are found in Guinea: *Acrostichum aureum*, *Avicennia germinans*, *Conocarpus erectus*, *Laguncularia racemosa*, *Rhizophora harrisonii*, *Rhizophora mangle*, and *Rhizophora racemosa*. The species distribution of woody mangroves varies by location. *Avicennia* spp. needs greater substrate stability, and is generally found in the shallows along the channels going inland, whereas *Rhizophora* spp and *Laguncularia* prefer convex banks prone to high sedimentation. In-situ surveys will help mapping the species distribution and selecting the most suitable for planting, if and where needed.

2. Capacity building:

The activities included in this component aim at building capacity amongst local communities and stakeholders, by training and empowering people in the participation and management of several project aspects. These include:

- training programmes for local communities on community-based Natural Resource Management system
- improving the capacity of project staff and government partners on Project Cycle Management

- extending awareness programmes amongst local communities on biodiversity and mangrove ecosystems
- coordinating implementation of livelihoods/ sustainable community projects with special focus on women

3. Mangrove restoration and conservation

- Establishment of local management and monitoring committees
- Regular monitoring of field activities, biodiversity, plant re-growth and surveillance
- Set up permanent sample plots for the carbon monitoring
- Identification of leakage areas

4. Community development

- Baseline survey of local communities on their livelihood systems based on Sustainable Development Goals (SDGs). Conducting socio-economic surveys with regular monitoring in the proposed area, to collect data on number of people in households, occupation, living condition, education level, income, fuel-wood consumption, commercial fish harvesting, etc. These data will serve the purpose of setting a baseline to monitor project impacts on livelihood standard of the community
- Formulating potential sustainable energy, fishery and agriculture practices for local community, including also sustainable cookstove and salt production
- Developing income generation activities to secure local livelihood improvements

5. Project management, monitoring and evaluation

- Day-to-day technical and financial project management
- Project plan implementation monitoring and adjustment
- Monitoring, verification and reporting

There is not currently any compliance market, such as an Emissions Trading System (ETS) or cap-and-trade system, nor any Article 6 Framework in force or under development in Guinea⁵.

The project will not operate in an area covered under a Jurisdictional REDD+ (Reducing Emissions from Deforestation and Forest Degradation in Developing Countries), as the Guinean government has not submitted any programme (REDD+ Web Platform accessed on the 3rd February 2026).

Guinea is a signatory to the Paris Agreement, and has presented its revised Nationally Determined Contribution in 2025 (3rd revision), which covers the forestry sector. Guinea has expressed its willingness to engage in the voluntary carbon market, but it has not provided specific information about how it plans to do so. Nonetheless, the Government of Guinea has committed to create a Carbon Market Office (Cellule

⁵ International Carbon Action, 2025 <https://icapcarbonaction.com/en/ets>

Nationale Carbone) which oversees projects, deals with Corresponding Adjustments and other requirements related to the Carbon Market (e.g. BTRs).

1.13. Project Location

The proposed project site is located along the northern coasts of Guinea, adjacent to the border with Guinea-Bissau. The project site lies within the administrative regions of Boké.

Blue Coast Carbon has identified two suitable areas for the implementation of a mangrove conservation project (Figure 1), which were selected based on the forest maturity and the level of deforestation threat.



Figure 1 Areas proposed for Blue Coast Carbon project implementation

Site 1 – North-Frontier and Site 2 – Centre-North are located in the north of the Country, next to the border with Guinea-Bissau, and south from Kamsar, respectively (Figure 2). These sites include the Marine Protected Areas of Iles Tristao and Rio Kapatchez, designated Ramsar Wetlands of International importance (see Section 1.1). Site 1 has a total of 62,683 ha of mangrove forests, while mangroves cover a total of 11,068 ha in Site 2.

In total, the proposed project sites include 73,751 ha of mangrove forests.



Figure 2 Sites proposed for mangrove conservation Lamantin project. From left to right: Site 1 (North-Frontier), Site 2 (Centre-North).

1.14. Conditions Prior to Project Initiation

Avoided mangrove deforestation:

Ecosystem type: Mangrove forests encompassing pristine mangrove areas and degraded/deforested mangrove areas along Guinea's coast over a coastal mud substrate.

Current and historical land use: The current and historical land use is mangrove forests with areas that have been degrading rapidly due to unsustainable overharvesting of mangrove wood.

Mangrove restoration:

Ecosystem type: Degraded/deforested mangrove forests, abandoned mangrove rice fields and areas used for solar salt no longer operated

Current and historical land use: The area used to be covered by mangrove forests that have been deforested for fuelwood and charcoal, especially for fish-smoking and salt production, and for mangrove rice cultivation. These are currently abandoned degraded areas.

Analysis of satellite imagery shows that targeted restoration areas have been cleared of mangroves more than 10 years from the project start date.

Environmental conditions of the project area

Topography: The coast of Guinea is approximately 300 km long and is featured by sand and mud, estuaries rich in mangroves, and a wide continental shelf favouring the amplification of the tides. The coast is

indented, with rias (drowned river valleys) that form inlets, tidal marshes, mangrove forests, and estuaries. The continental shelf of Guinea extends 300 km along the coast, and covers an area of 47,400 km², making it the largest continental shelf in West Africa⁶.

Climate: The West African coast has a subtropical to tropical climate, primarily influenced by the Atlantic Ocean and the Sahara Desert⁷. The West African monsoon, a seasonal low-level atmospheric circulation pattern, represents, along with the Harmattan system, the primary driver of climate variability in the region. The coasts of Guinea are located within the Guineo-Congolian bioclimatic region, which is the wettest in West Africa, with average annual rainfall between 2,200 and 5,000 mm. The rainfall can be distributed across most of the year, or in two rainy seasons with short drier periods between the rains.

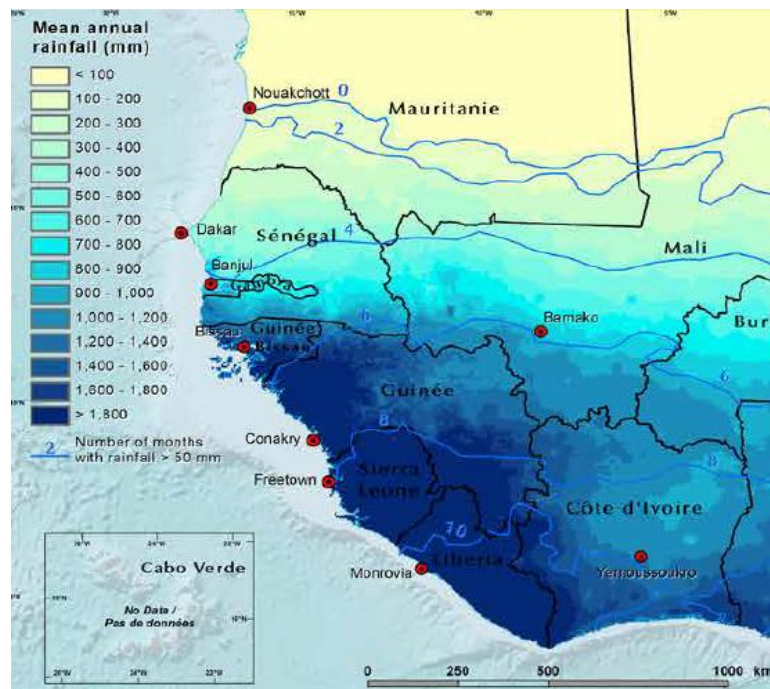


Figure 3 Mean annual rainfall 1981–2014, with the number of months of 50 mm or more of rainfall (Adapted from CILSS, 2016).

Hydrology & hydrodynamics: The most important rivers originate in the Guinea highlands and include the Niger, Volta, and Senegal rivers⁸⁹. For hydroelectric power generation, irrigation, and flood control purposes, most of these large rivers as well as smaller ones have been dammed in recent decades,

⁶ UNEP. 2007. "Mangroves of Western and Central Africa." UNEP-Regional Seas Programme/UNEP-WCMC.

⁷ Lewis K, Buontempo C (2016) Climate impacts in the Sahel and West Africa: the role of climate science in policy making", <https://doi.org/10.1787/5f1smktwcd0-enresilient> coasts (worldbank.org)

⁸ Diop S, Fabres J, Pravettoni R, Barousseau JP, Descamps C, Ducrot JP (2014) The western and central africa land-sea interface: a vulnerable, threatened, and important coastal zone within a changing environment. In: The land/ocean interactions in the coastal zone of west and Central Africa (pp. 1–8). Springer, Cham

⁹ Anthony EJ (2015) Patterns of sand spit development and their management implications on deltaic, drift aligned coasts: the cases of the Senegal and Volta River delta spits, West Africa. In: Randazzo G, Cooper JAG (eds) Sand and gravel spits coastal research library series 12. Springer, Berlin, pp 21–36

significantly altering their hydrology and sediment flows, accelerating coastal erosion downstream⁸⁹¹⁰. The West African coast is generally considered a storm-free environment, dominated by North Atlantic distant swells¹¹¹² with their interannual fluctuation described by the North Atlantic Oscillation⁸. The wave regime in the Gulf of Guinea area is mainly driven by extratropical storms, and their variability is modulated by the South Atlantic Annular Mode (SAM)¹³¹⁴. The evolution of the sandy coast is controlled by strong longshore sediment drift resulting from oblique waves¹¹¹⁵. Tidal amplitude is typically micro- to meso-tidal, with the highest amplitudes recorded in Guinea, Guinea-Bissau and Sierra Leone (2.8–4.7 m)⁸.

Biodiversity, ecosystems & vegetation: The richness of estuaries, deltas, coastal lagoons, and nutrient-rich cold water upwelling contributes greatly to the biodiversity in West Africa. These habitats include (1) wetlands, particularly those surrounding mangroves in a succession of deltas and estuaries, providing key habitats for migratory birds¹⁶; (2) coastal freshwater lagoons, deltas and estuaries, and a wide range of tidal marshes and seasonal swamps, also counting with seagrass beds located off some estuaries, archipelagos and deltas; (3) sandy beaches, considered important nesting ecosystems, especially for sea turtles. The dominant factor affecting the waters of the continental shelf is coastal upwelling, which is the source of the rich fish production of the region.

Guinea's coastal and marine areas are characterised by the presence of still-preserved ecosystems, including large mangrove zones and rich natural-resource waters that are habitats for important plant and animal species, including the endangered West African manatee, pygmy hippos, a variety of Cetacea, and critically endangered migratory birds.

In western Guinea, two ecoregions characterize some 300 km of Atlantic coastline and coastal plains — the Plaines Côtières (PC – Coastal Plains) and Zones de mangroves (ZM – Mangrove Zones).

10 Abessolo GO, Larson M, Almar R (2021) Modeling the Bight of Benin (Gulf of Guinea, West Africa) coastline response to natural and anthropogenic forcing. *Regional Stud Marine Sci* 48:101995

11 Almar R, Kestenare E, Boucharel J (2019) On the key influence of remote climate variability from Tropical Cyclones, North and South Atlantic mid-latitude storms on the Senegalese coast (West Africa). *Environ Res Commun* 1:071001

12 Ndour A, et al. (2020) On the natural and anthropogenic drivers of the Senegalese (West Africa) low coast evolution: Saint Louis Beach 2016 COASTVAR experiment and 3D modeling of short term coastal protection measures. In: Malvárez, G. and Navas, F. (eds.), *Global Coastal Issues of 2020*. J Coastal Res, Special Issue No. 95, pp. 583–587. Coconut Creek (Florida), ISSN 0749–0208.

¹³ Almar R, et al. (2015) Response of the Bight of Benin (Gulf of Guinea, West Africa) coastline to anthropogenic and natural forcing. Part 1: Wave Climate variability and impacts on the longshore sediment transport. *J Cont Shelf Res* 110:48–59

¹⁴ Abessolo GO, Almar R, Jouanno J, Bonou F, Castelle B, Larson M (2020) Beach adaptation to intraseasonal sea level changes. *Environ Res Commun* 2:051003

¹⁵ Anthony EJ, Almar R, Besset M et al (2019) Response of the Bight of Benin (Gulf of Guinea, West Africa) coastline to anthropogenic and natural forcing, Part 2: sources and patterns of sediment supply, sediment cells, and recent shoreline change. *Cont Shelf Res* 173:93–103

¹⁶ Diop S, Fabres J, Pravettoni R, Barousseau JP, Descamps C, Ducrotoy JP (2014) The western and central africa land–sea interface: a vulnerable, threatened, and important coastal zone within a changing environment. In: *The land/ocean interactions in the coastal zone of west and Central Africa* (pp. 1–8). Springer, Cham

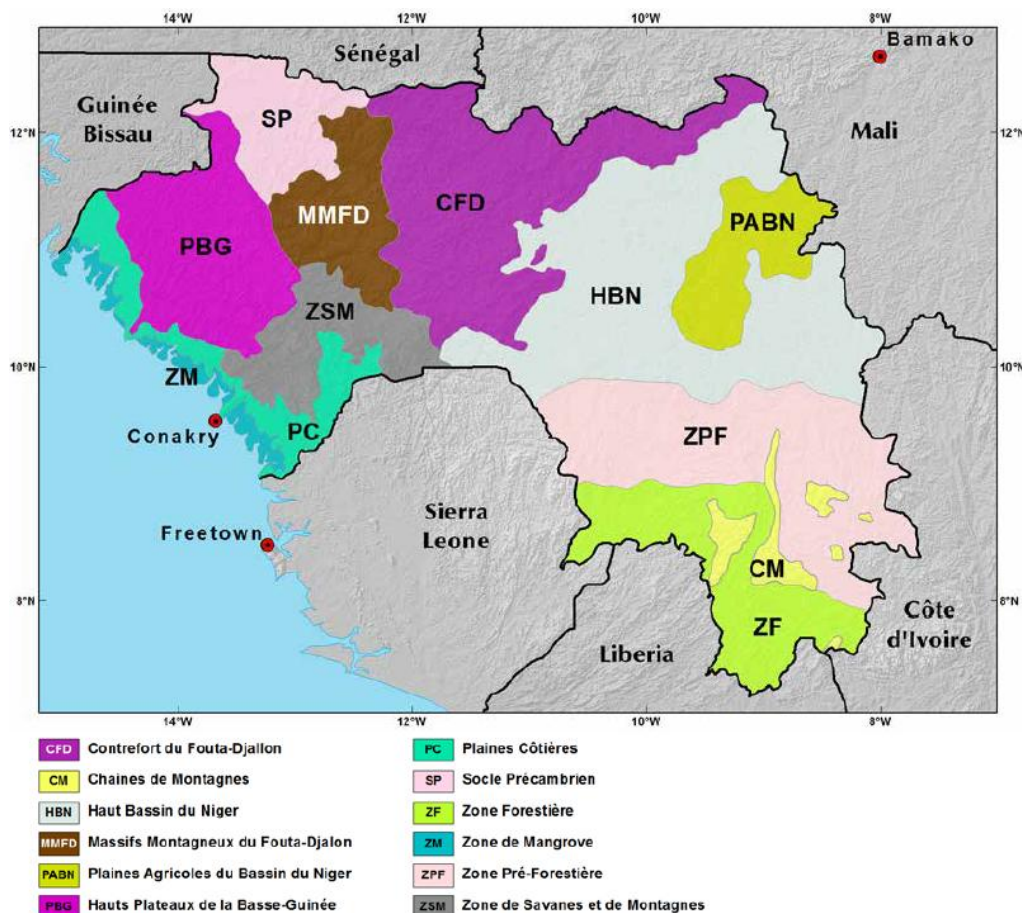


Figure 4 Ecoregions of Guinea (Source: CILSS, 2016).

Mangroves are found all along the Guinean coast except for Cape Verga and Kaloum Island. The topography of the coastal area facilitates the deposition of sediment and submersion of the mouths of the rivers. There is a long tidal reach up the estuaries, which causes flooding of the rivers, leaving raised bars. It is here that mangroves can develop, within the bay of the estuary. Mangroves extend more than 10 km inland and, along the widest rivers, they can be found up to 40 km inland from the coast¹⁷.

The maximum possible area suitable for mangroves in Guinea, based on geomorphology, is estimated to be around 4,000 km² ¹⁸. Of these, approximately 1,300 m² of land currently do not have any mangrove cover because these areas have been cleared for agriculture, salt production, or for increased urbanization, including the building of ports. Others are simply lying abandoned. The remaining 2,700 km² currently under mangrove cover also faces various levels of degradation.

¹⁷ Profile, C. 2005. "Global Forest Resources Assessment 2005: Thematic Study on Mangroves."

¹⁸ Based on the conversion of an estimated 1,400 km² of mangrove swamp being converted to rice fields, of which 620 km² lay abandoned in 1993. (Spaulding 1997)

Common mangrove species found in Guinea: Seven species of mangrove are found in Guinea: *Acrostichum aureum*, *Avicennia germinans*, *Conocarpus erectus*, *Laguncularia racemosa*, *Rhizophora harrisonii*, *Rhizophora mangle*, and *Rhizophora racemosa*. The species distribution of woody mangroves varies by location. *Avicennia* needs greater substrate stability, and is generally found in the shallows along the channels going inland, whereas *Rhizophora* and *Laguncularia* prefer convex banks prone to high sedimentation.

All Guinean species are listed as ‘least concern’ on the IUCN Red List of threatened species¹⁹.

The non-woody mangroves of Guinea tend to be located in degraded areas²⁰. In denuded areas, *Avicennia* and *Rhizophora* are found in mixed settlements along the banks of the channels. More developed “forest cathedrals” are found along the Konkouré River where there is fresh water from inland. *Avicennia germinans* and grassy species often colonize the deforested areas intended for rice growing. *Rhizophora racemosa* can reach 25 m in height in Kakounsou and in the Bay of Sangaréya, but in other areas the trees seldom exceed 8 m and are often much smaller. In the same zone, *Avicennia germinans* grows up to 15 m, and *Conocarpus erectus* can also be found here²¹.

Mangrove species	Characteristics	Occurrence	Propagation
Avicennia germinans (Black Mangrove)	Develops pneumatophores (aerial roots), which protrude from the soil around the tree’s trunk.	Occurs at higher elevations inland than the red mangrove.	Reproduces by vivipary, with sprouting seeds that drop into the soft bottom around the base of the trees. The seeds can also be transported by currents and tides to other suitable locations.
Laguncularia racemosa (White Mangrove)	Does not develop visible pneumatophores, and has elliptical light yellow-green leaves.	Occurs at even higher elevations farther upland than either the red or black mangroves.	Sprouting seeds that drop into the soft bottom around the base of the trees. The seeds can also be transported by currents and tides to other suitable locations.

¹⁹ <https://www.iucnredlist.org/>

²⁰ UNEP. 2007. “Mangroves of Western and Central Africa.” UNEP-Regional Seas Programme/UNEP-WCMC.

²¹ Profile, C. 2005. “Global Forest Resources Assessment 2005: Thematic Study on Mangroves.”

<i>Conocarpus erectus</i> (Silver-leaved Buttonwood)	Belongs to the same family as the white mangrove, but has a different appearance, with dense, rounded flowerheads that grow in a branched cluster, and purplish-green, round, conelike fruit.	Grows in brackish areas and alkaline soils, thriving in the broken shade and wet soils.	The seed heads burst when ripe, and the seeds are dispersed by water.
<i>Acrostichum aureum</i> (Golden Leather Fern)	Does not have any pneumatophores; instead has fibrous, fernlike roots. It can grow up to 1.5 m high and have leaves that can be up to 1 m long and 4 cm wide.	Grows in swamps and mangrove forests, salt marshes, and on river banks; is tolerant of raised salinity levels.	Mature fronds become sporophyllous, diffuse sporangia at the abaxial surface, mixed sporangia on both sides of the mid-vein, brown sporangia stalked, upper globose.
<i>Rhizophora harrisonii</i>	A hybrid of <i>R. mangle</i> and <i>R. racemosa</i> ; shares morphological characteristics with both species and is recognized by FAO as a distinct species.	Mostly found in the estuaries of river systems with more continuous freshwater flows.	Creates a propagule that is a fully-grown propagule on the mangrove and is capable of rooting and producing a new tree.
<i>Rhizophora mangle</i> (Red Mangrove)	Usually grows to 10–20 m in height and has a 10–30 cm diameter at breast height (DBH); in some sites has been observed to exceed 40 m in height and 70 cm DBH.	Generally found growing along the water's edge.	Sprouting seeds that drop into the soft bottom around the base of the trees. The seeds can also be transported by currents and tides to other suitable locations.
<i>Rhizophora racemosa</i>	Grows up to 30 m tall, often with aerial stilt roots; in more marginal habitats is shorter, more branched, and	It is a pioneering species, being the first mangrove to appear in mud flats. It primarily occurs in the open	The fruit produces propagules that may fall into the water and be

	scrubby. The leaves grow in opposite pairs, each pair with two interlocking stipules. The leaves are simple and entire, with elliptical hairless blades and slightly down-rolled margins.	lagoon systems in areas along the coastlines.	dispersed by wind and currents.
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One of the specificities of the mangrove areas of Guinea is the existence of three specific types of plains:

- **Estuarine plains.** These border the estuaries of the mouths of coastal rivers. They are very submersible, and are naturally occupied by *Rhizophora* mudflats (which are dominant).
- **The seafront plains.** These are located in sectors further from the mouths of rivers. They are much less submersible than the estuarine plains, but with salty soils. They were originally occupied by *Avicennia*; since the 1950s they have been used for rice cultivation.
- **The back mangrove plains.** These are floodable (especially during the rainy seasons). They are found in the transition between continental (non-salty) brackish and salty environments, and are partially colonized by hydrophytes with drainage constraints.

Mangroves in Guinea provide habitats and feeding grounds for several migratory shorebird species, offshore fisheries, and important marine species. The common bird species include the African spoonbill (*Platalea alba*), common pratincole (*Glareola nuchaltis*), the near threatened curlew (*Numenius arquata*) and the vulnerable Rufus fishing owl (*Scotopelia ussheri*)²²²³.

Vertebrate faunae include the threatened West African manatee (*Trichechus senegalensis*), the Water Chevrotain (*Hyemoschus aquaticus*) and the common Bottlenose dolphin (*Tursiops truncatus*). The Konkouré and Tristao Ramsar sites are home to the vulnerable African dwarf crocodile (*Osteolaemus tetraspis*), the Nile crocodile (*Crocodylus niloticus*), and the African sharp-nosed crocodile (*Mecistops cataphractus*). Other reptile species include the common Senegal mabuya (*Trachylepis affinis*), the rare Nimba snake-eyed skink (*Panaspis tristaoi*), the half-toed gecko (*Hemidactylus albivertebralis*) and the African rock python (*Python sebae*).

Primates including the endangered Red Colobus monkey (*Procolobus badius*), the small green monkey (*Cercopithecus sabeus*) and the common patas monkey (*Erythrocebus patas*) can also be spotted in the mangroves of Guinea. Guinea's beaches are breeding grounds for leatherback (*Dermochelys coriacea*), loggerhead (*Caretta caretta*), green (*Chelonia mydas*), and olive ridley (*Lepidochelys olivacea*) sea turtles

²² Thiollay, JM. Important Bird Areas in Africa and Associated Islands. Priority Sites for Conservation. 2001. Biodiversity and Conservation 11, 1697–1698 (2002). <https://doi.org/10.1023/A:1016864606424>

²³ Guinea - BirdLife Country Profile, accessed on the 29th September 2025 <https://datazone.birdlife.org/country/factsheet/guinea>

and the mangroves provide important nursery areas for juvenile sea turtles and fish. Fish species of commercial importance found in Guinea's mangroves include the *Tilapia* spp, the endangered Senegalese otolith (*Pseudotolithus senegalensis*), the near threatened Sea banana (*Albula vulpes*), and mullets (*Mugil cephalus*), shrimp (*Palaemon hastatus*) and crabs (*Portunus validus*)²⁴.

Land-use pressures on mangrove forests: The country's coastal areas are subject to a number of environmental impacts arising from population growth and development within the country's key economic sectors (agriculture, fisheries, mining, and infrastructure). Rapid population growth has resulted in increased exploitation of coastal resources like mangroves and fisheries, many of which raise issues of sustainability.

The most common use of mangroves in Guinea is as fuelwood and charcoal, especially for fish-smoking and salt production. Mangrove wood is also used for construction purposes, including the roofing of houses. Currently the conversion of mangrove wetlands for salt production, and the expansion of towns and villages are major destructive threats to mangroves in Guinea²⁵.

In West Africa, including Guinea, salt is either produced by cooking (i.e. evaporation using heat generated by firewood) or by sun-dried crystallization. The cooked salt consumes a considerable amount of wood from the mangrove while the sun-dried salt is less productive in the local communities. In Guinea, it is estimated that the production of 1 kg of salt requires 3.1 kg mangrove wood²⁶.

To produce salt, the mangrove vegetation is removed in selected areas and the underlying soils are dug up to a depth of around 10 cm to facilitate evaporation. After a few days, the upper crusts are raked and scraped and, finally, collected into heaps for processing. These soils are leached several times with the same water and the saline solution is then boiled to obtain crystallized salt. The firewood requirement is met by the neighbouring mangroves. The environmental consequences of such salt production include the formation of barren depressions in which the hydrological regime is modified, destruction of mangroves for firewood, compaction of the surface soil by repeated raking and trampling, and the creation of ecological conditions unsuitable for the recolonization of mangroves.

Current estimates indicate that approximately 220,000 tonnes of mangrove cuttings is in direct use each year for salt manufacturing, fish curing, and as fuelwood and timber in Conakry and in rural areas of the country. However, the annual natural mangrove wood growth per year in the 1,000 km² of commercially suitable area is only 80,000–150,000 tonnes, depending on the type of mangroves. This clearly demonstrates a reason for the loss of mangrove cover from overexploitation. However, it also shows that if

²⁴ Beth A. Polidoro et al., "The Status of Marine Biodiversity in the Eastern Central Atlantic (West and Central Africa)," *Aquatic Conservation: Marine and Freshwater Ecosystems* 27, no. 5 (October 2017): 1021–34, <https://doi.org/10.1002/aqc.2744>

²⁵ International Bank for Reconstruction and Development (2023). *West Africa Coastal Areas High-Level Platform Benefits of Mangrove Afforestation and Restoration Programs in Guinea-Conakry*.

²⁶ Siddighi, Balde & Kobayashi, Hajime & Nohmi, Makoto & Ishida, Akira & Matsumura, Ichizen & Esham, Mohamed & Tolno, Emmanuel. (2013). *SOCIO-ECONOMIC ANALYSIS OF SMALL-SCALE SALT PRODUCTION TECHNIQUES IN THE COASTAL AREA OF GUINEA: AS AN ALTERNATIVE FOR IMPROVING LIVELIHOOD STATUS AND SUSTAINABLE MANGROVE FOREST MANAGEMENT*. *International Journal of Research in Engineering, IT and Social Sciences*. Volume 3. 1-23.

all 2,700 km² can be made commercially suitable, the annual sustainable yield would range from 210,000–410,000 tonnes, and could help to arrest the degradation of mangrove areas.

The clearing of mangroves for fuelwood and charcoal, timber, and salt extraction has resulted in mangrove degradation, and expansion of the port of Kamsar in the Bay of Sangaréya has resulted in the loss of 700,000 m² of mangroves. Fortunately, reforestation of the same area using the *Rhizophora* and *Avicennia* species was undertaken between 1993 and 1998 to help mitigate these effects²⁷.

The extractive industry is also increasingly space-consuming in Guinea. The extraction of bauxite, an ore needed to manufacture aluminium, is on the rise since 2014²⁸. In 2020, Guinea was the world's second-largest producer of bauxite and has the largest reserves in the world (25%)²⁹. Most of the production (80%) is located in the Boké region in the north of the Country. The processing and transport of bauxite that takes place in the coastal zone has also affected the coastal ecosystem, although the extent of the impact has not been studied extensively.

Losses of mangrove areas have also been caused by the reclamation of land for agriculture. Mangrove rice cultivation is widespread in Guinea. It is made possible by mangrove bunds that protect the land from flooding. Most of the land used for rice cultivation has been obtained over time by clearing erstwhile mangrove plantations.

Twenty-three percent of the national rice production in Guinea is from mangrove areas, and it provides livelihoods for more than 50,000 rice farmers, the majority of whom are women. Shifting to agriculture also occurs where salt inundation is low. Mangrove rice is a flood-fed crop requiring two flooding cycles, one with freshwater and one with saltwater. The saltwater is needed to eliminate weeds, regulate soil pH, and deposit organic matter and nutrients. Freshwater is necessary for soil desalination prior to cultivation, and maintenance of the water table throughout the cycle.

In ideal conditions, mangrove rice farming produces higher yields than other rice farming techniques used in the interior of the country. Annual mangrove rice production averages around 376,000 tonnes; it represents 16% of rice farmland in the country, and 23% of national production³⁰.

Of an estimated 1,400 km² of mangrove that had been converted to rice fields in Guinea by 1993, 620 km² had been abandoned³¹. The accelerated rate of conversion of mangrove forests to rice in Guinea has resulted in irreversible damage to these forests.

Sea level rise & erosion: In addition to mangroves being cut to support subsistence rural livelihoods and conversion to rice fields, sea-level rise and other climatic phenomena are also contributing toward

²⁷ UNEP. 2007. "Mangroves of Western and Central Africa." UNEP-Regional Seas Programme/UNEP-WCMC.

²⁸ Initiative pour la transparence dans les industries extractives en Guinée. Rapport Assoupli 2019-2020. <https://www.itie-guinee.org/rapport-assoupli-exercices-2019-2020/>

²⁹ <https://pubs.usgs.gov/periodicals/mcs2021/mcs2021-bauxite-alumina.pdf>

³⁰ Agence Française de Développement 2016

³¹ Spalding M. , F. Biasco, C Field 1997. Word Mangrove Atlas. International Society or Mangrove Ecosystems. Okinawa, Japan.

mangrove degradation in Guinea. These occur through the loss of coastal and intertidal habitats, including changes in geomorphological processes, and the increased vulnerability of woody formations (UNDP 2019). Studies also reveal that sea-level rise is a major potential threat to mangrove ecosystems due to their sensitivity to the duration and frequency of flooding, and salinity levels that may exceed the physiological threshold of mangrove-specific tolerance. Additionally, the failure to maintain dikes built in previous decades, and the lack of planning and integration of climate risks in construction activities on the coast have contributed to the coastal zone's vulnerability to erosion, and have affected the mangroves. Sea-level rise will increase the direct inundation of low-lying areas, facilitate rapid erosion of the soft shores, and increase offshore loss of sediment as well as flooding.

The projected total sea level is expected to raise by 0.78 m by 2100, based on the sea level projections relative to the period 1995-2014 for the West African region for Shared Socioeconomic Pathway (SSP) scenario SSP5-8.5 (very high GHG emissions), as defined by the 6th Assessment Report from the IPCC³².

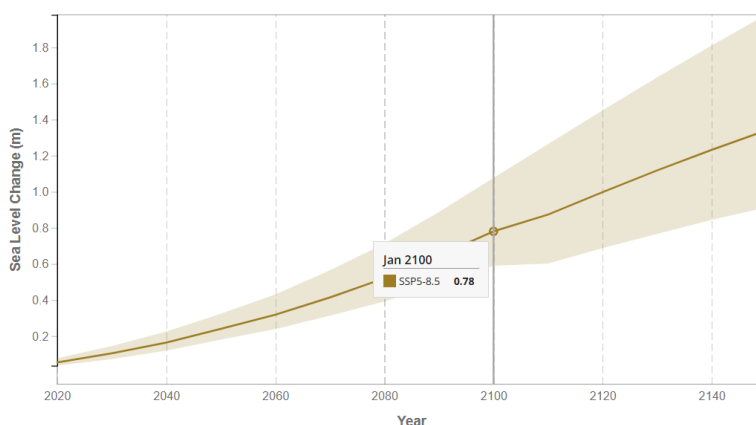


Figure 5 Sea level projections relative to the period 1995-2014 for the West African region³².

It is estimated that a 1 m increase in sea level will inundate a significant portion of the coastal areas, causing major problems for mangrove rice cultivation.

Mangrove deforestation trends: Estimates show that Guinea currently has a little over 2,000 km² of mangrove forests, down from 4,000 in 1957³³³⁴. Between 1980 and 2006, it is estimated that the mangrove area in Guinea fell from 2,992 to 2,039 km², representing a loss of 32 %²⁷³⁵. A 2019 survey by NASA estimated the mangrove area to be 2,076 km².

³² Data accessed through [the NASA Sea Level Projection Tool](#)

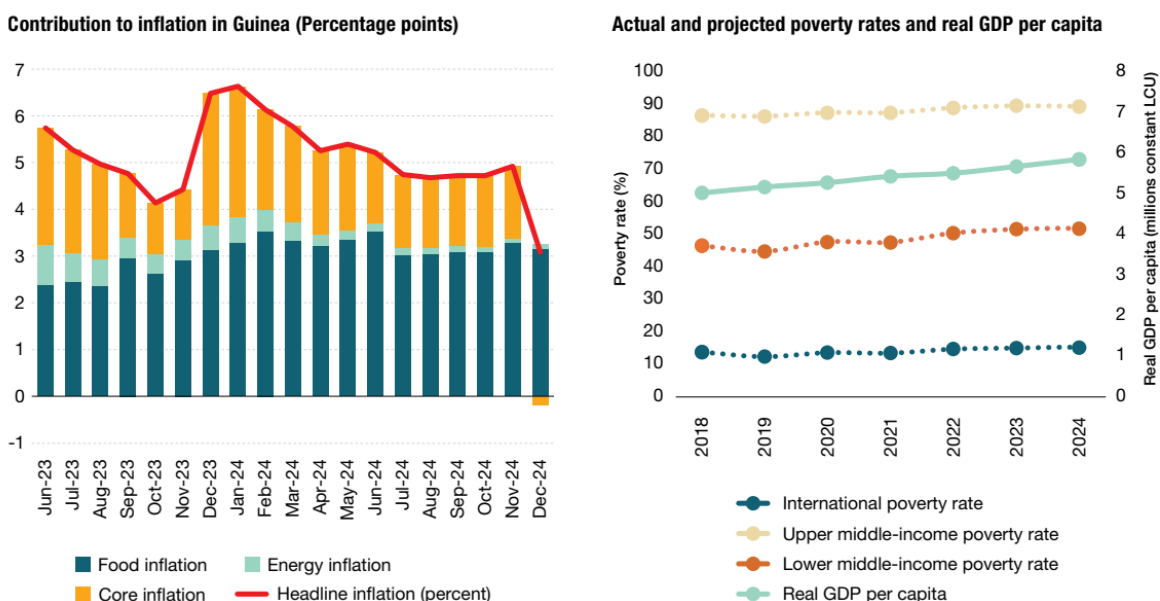
³³ Diallo M. et al. 2019: Model of Criteria Taking into Account Halieutic Biodiversity in Strategic Port Planning in Guinea." *Vertigo: The electronic journal in environmental sciences*, vol.19 n° 3.

³⁴ Rouanet 1957, cited in Kaba, B. 2001

³⁵ Ajonina, G., A. Diamé, and J. Kairo. 2008. "Current Status and Conservation of Mangroves in Africa: An Overview." *World Rainforest Movement Bulletin* 133: 1-6.

According to the Global Mangrove Watch, in 2020 the area of mangrove habitat in Guinea was 2,211.45 km², with a deforestation rate between 2010 and 2020 of 1.9%³⁶. Estimates reveal that only about 1,000 km² of this total area is in a good enough state that it could allow for commercial exploitation of mangroves if done on a sustainable basis³⁷.

Socio-economic situation: Despite its abundant natural resources, growing population, and privileged geographic location, 52% of Guineans live below the international poverty line of \$3.65 a-day, with the poverty rate increasing by 7% between 2019 and 2024³⁸. This limited poverty reduction is due to weak productivity growth resulting from low levels of physical and human capital, declining competitiveness of non-mining sectors, and significant structural inequalities in income, education, and infrastructure.



Source: World Bank, National Institute of Statistics.

Figure 6 Contribution to inflation in Guinea (left) and poverty rates and real GDP per capita (right)³⁶.

Food inflation is elevated and is the main driver of inflation in Guinea, contributing 3.3% to the national annual inflation rate of 5.1% in 2024. The increase in food prices disproportionately affects the poor, with low-income populations being particularly vulnerable as they spend a larger share of their income on food. Persistent food inflation could exacerbate food insecurity in Guinea where the prevalence of severe food insecurity among the total population is already high, estimated at 50%. In fact, despite real GDP per capita increasing by 3.1% in 2024, the poverty rate (<\$3.65-a-day in 2017) increased by 0.3ppts to 52% due to slightly higher annual inflation and insufficient non-mining sector job creation³⁸.

³⁶ www.globalmangrovetwatch.org (accessed on 6th November 2025)

³⁷ RUE O. (1998): The development of the coast of Guinea - Memories of mangroves - ed. The Harmattan

³⁸ World Bank. 2025. Guinea Economic Update: Domestic Resource Mobilization and Management for Inclusive and Sustainable Development, DC: World Bank.

Female-headed households are more vulnerable to poverty due to lower educational attainment. Women aged 25-35 are 19% more likely to live in poverty compared to men (43.4% vs. 24.4%). This is exacerbated by the school enrolment gap for girls, which widens at the secondary level. The lack of investment in women's human capital increases their poverty risk and limits their economic participation. Limited intergenerational mobility in education perpetuates poverty across generations. Nationally, 73% of adults whose fathers lacked formal education also lack formal education, with the share rising to 84% in rural areas. This suggests persistent income disparities and inequality unless access to education is expanded³⁸. Women living in coastal areas are heavily dependent on mangrove products, further highlighting their vulnerability³⁹.

1.15. Compliance with Laws, Statutes and Other Regulatory Frameworks

The project will be in line with all laws and statutory requirements of Guinea such as the Civil Code (2019) and the ones regarding :

Labour: Guinea Labor Code (Loi N°L/2014/072/CNT portant Code du travail) especially the articles about working conditions and presidential decree of 1st June 2022 regarding the new Guinean Minimum Salary;

Land ownership: Land law in Guinea is governed by the following texts:

- Constitution of 2010, which guarantees its citizens the right to private property.
- The Guinea Land Code of 1992 (*Code Foncier et Domanial*), which stipulates that the owner is the person who provides proof of the occupation of a plot of land through its use (creation of value) or through a public inquiry⁴⁰. This law establishes access to individual property for individuals and legal entities, while maintaining the presumption of State ownership of vacant and unowned land. This reform also creates land commissions in each prefecture of Guinea⁴¹. This law was essentially designed to address urban specificities: only Article 92 addresses the issue of rural land tenure, albeit in a very vague manner.
- The Rural Land Policy of 2001 (Déclaration de la Politique Foncière en Milieu Rural) acknowledges specific customary land rights and encourages the creation of laws to formalize these rights with an increase of local participation. However, this policy was not accompanied by implementation measures.

39 WACA (2023). Technical Report: Understanding the Nexus of Mangroves and Women in Guinea-Conakry. <https://www.wacaprogram.org/knowledge/technical-report-understanding-nexus-mangroves-and-women-guinea-conakry>

40 Article 39: The following are owners within the meaning of this Code: 1° - natural or legal persons holding a land title; 2° - occupants, natural or legal persons, holders of a land register, residence permit or occupation permit; 3° - occupants, natural or legal persons, who can prove that they have occupied a building peacefully, personally, continuously and in good faith and as owner. If necessary, proof of good faith shall be provided by all means, in particular by payment of property taxes relating to the said building, by the development of the building in accordance with local customs or by a public and adversarial inquiry.

41 Diop, Moustapha. 2017. Réformes foncières et gestion des ressources naturelles en Guinée : enjeux de patrimonialité et de propriété dans le Timbi au Fouta Djallon. Paris: Karthala. <https://landportal.org/library/resources/9782845868564/r%C3%A9formes-fonci%C3%A8res-et-gestion-des-ressources-naturelles-en-guin%C3%A9e>.

The process of revising the land laws, which began in 2012, has still not been completed 10 years after its launch⁴².

- The Pastoral Rights Law of 1995 (Loi L/95/51/CTRN portant Code Pastoral) provides herders with open and unrestricted access to pastoral areas and the natural resources within them, with the condition of avoiding overuse and respecting the customary rights of other users.

Forests and restoration:

- Guinea's Forestry Code (Loi ordinaire U2017/060/AN du 12 décembre 2017, portant code forestier de la République de Guinée) especially articles 51, 67, 124, 128, 137 and 142 which encourage forest restoration.

- Guinea's Environmental Code (Decret D/2019/221/PRG/SGG portant promulgation de la loi L/2019/0034/AN du 04 juillet 2019) especially article 38 and 39.

-Wildlife Protection Code and Hunting Regulations (Loi Ordinaire N° 2018/0049/AN Portant Code de protection de la faune sauvage et réglementation de la chasse and Arrêté A/2020/1591/MEEF/CAAB/SGG Portant protection des espèces de faune et de flore sauvages en République de Guinée).

Water:

- Code Maritime regulates the maritime waters under the jurisdiction or sovereignty of the Republic of Guinea, vessels registered in the Republic of Guinea, their crews, and passengers on board, the crews and passengers of foreign vessels, and floating platforms located in waters under national jurisdiction.

- Code de la Pêche maritime, regulating maritime fishing in the Republic of Guinea, defines the rules applicable to fishing in the maritime zones under Guinean sovereignty or jurisdiction, as well as the rules applicable to Guinean fishing vessels operating beyond these zones.

- Code de l'Eau regulates water resources, including a set of measures to be taken to ensure their permanent quantitative and qualitative inventory, protection, enhancement, and optimal use, taking into account the social, economic, and cultural needs of the nation.

In addition, Guinea is a signatory to the African Forest Landscape Restoration Initiative (AFR100) which is a pan-African effort aimed at restoring 100 million hectares of deforested and degraded landscapes across the African continent by 2030. This ambitious initiative seeks to address critical issues such as land degradation, climate change mitigation, and biodiversity conservation. AFR100 brings together a diverse range of stakeholders, including governments, local communities, non-governmental organizations, and

⁴² <https://data.landportal.org/book/narratives/2022/guinea#ref7>

private sector partners, to collaborate on large-scale restoration projects. Within this framework, Guinea aims to restore 55% of degraded land and restrict the loss of undegraded land to 1% before 2030⁴³.

There is currently no specific legislation for mangrove forests in Guinea. However, the importance of mangroves for the climate, biodiversity, and society is recognized by state actors, and policy frameworks are being developed at the national and regional levels for this habitat.

The Ministry of Environment and Sustainable Development is responsible for:

- developing the Mangrove Management Master Plan;
- consulting with communities in mangrove areas affected by high deforestation rates;
- proposing solutions to mangrove degradation.

However, Guinea has not yet established dedicated coastal protected areas, and mangrove forests are only partially included in protected areas.

Other weaknesses related to the protection of mangrove forests include a limited capacity of government institutions to reform, weak academic training, and a lack of resources to monitor and enforce regulations on the ground.

Furthermore, the laws governing protected areas are sometimes contradictory. For example, the Ministry of the Environment is legally responsible for the management of classified forests, but in practice, the prefect is also authorized to grant occupancy permits to individuals, without taking mangroves into account in the absence of a clear legal framework.

Protected areas are poorly defined; they include classified forests, national parks, etc., but this definition is not unanimous, and a protected areas code is currently being finalized.

The proposed activities fall within a regulatory framework that is evolving favorably, but conservation frameworks still lack rigorous enforcement in coastal areas, demonstrating the regulatory overhang of the project's activities.

⁴³ <https://afr100.org/country/guinea#section-projects>

1.16. Double Counting and Participation under Other GHG Programs

1.16.1. No Double Issuance

Is the project receiving or seeking credit for reductions and removals from a project activity under another GHG program?

☐ Yes ☒ No

1.16.2. Registration in Other GHG Programs

Has the project registered under any other GHG programs?

☐ Yes ☒ No

Is the project active under the other program?

☐ Yes ☒ No

1.16.3. Projects Rejected by Other GHG Programs

Has the project been rejected by any other GHG programs?

☐ Yes ☒ No

1.17. Double Claiming, Other Forms of Credit, and Scope 3 Emissions

1.17.1. No Double Claiming with Emissions Trading Programs or Binding Emission Limits

Are project reductions and removals or project activities also included in an emissions trading program or binding emission limit? See the VCS *Program Definitions* for definitions of emissions trading program and binding emission limit.

☐ Yes ☒ No

1.17.2. No Double Claiming with Other Forms of Environmental Credit

Has the project activity sought, received, or is planning to receive credit from another GHG-related environmental credit system? See the VCS Program Definitions for definition of GHG-related environmental credit system.

☐ Yes ☒ No

1.17.3. Supply Chain (Scope 3) Emissions

Do the project activities specified in Section 1.12 affect the emissions footprint of any product(s) (goods or services) that are part of a supply chain?

☐ Yes

☒ No

1.18. Sustainable Development Contributions

The proposed project will contribute to the achievement of the following five sustainable development goals (SDGs) of the United Nations' Agenda 2030:

SDG 5: Gender equality (medium priority)

Monitoring, restoration, and surveillance activities will be carried out directly employing local communities for the whole duration of the project. Men and women will be involved in equal measure in all project activities, including stakeholder consultations, ensuring women's full and effective participation and equal opportunities at all levels.

This will be monitored by assessing the proportion of work positions held by women in project implementation and stakeholders consultation, including managerial positions.

The project will oversee the development of side economic activities owned by women, e.g. beekeeping, aquaculture projects, ecotourism.

SDG 8: Decent work and economic growth (medium priority)

The project will provide direct employment as well as development and entrepreneurship opportunities for local communities, including through training and professional development, to ensure long-term impact. The project is also designed to provide alternative livelihoods for the local community by testing and generating income streams from the project area through carbon, agroforestry, and other economic activities. Community initiatives will increase incomes and promote sustainable agricultural and fishing practices to improve living conditions. The protection, restoration and reduction of pollution in mangroves will also help to enhance food security by protecting fish nurseries and improving them for future fish stocks.

SDG 13: Climate action

The project will restore degraded mangrove forests, aiming at contributing to climate change mitigation by removing greenhouse gases from the atmosphere through mangrove forest restoration and planting. The new mangrove growth will enhance carbon stocks in biomass and soil. Restoring degraded ecosystems will capture more than 1,900,000 tonnes of CO₂e.

Mangroves provide long-term coastal protection from extreme weather and climate-related events, as they mitigate erosion and stabilize soil. Their intricate and expansive crown-root systems entrap sediments and dissipate turbulent kinetic energy which also enhances sediment deposition. Mangroves are also crucial in mitigating the impact of tropical cyclones and storm surges, and offer substantial protection against wind

damage during storms. Restoring mangroves will thus strengthen coastal resilience to sea level rise and more intense storms.

SDG 14 Life below water (low priority)

This project aims at establishing a new coastal protected area managed by an ecosystem-based approach, thus contributing to the achievement of SDG 14. The project area covers threatened coastal habitat that is under pressure from encroachment, development and pollution.

Furthermore, the high productivity and physical complexity of mangrove ecosystems make them important nursery grounds for many commercial and non-commercial fish and invertebrate species. The restoration and conservation activities foreseen will thus contribute to the restoration of fish stocks in Guinea. The project's restoration activities will promote the recovery of the natural ecosystem, species richness, and contribute to the reduction of nutrient pollution, thanks to the natural capability of mangroves to filter organic pollution.

SDG 15: Life on land (deforestation) (low priority)

This project will promote the implementation of sustainable management of mangrove forests, halt deforestation, restore degraded forests and increase reforestation in Guinea.

1.19. Additional Information Relevant to the Project

1.19.1. Leakage Management

Section to be completed in the PD submitted for validation.

1.19.2. Commercially Sensitive Information

No commercially sensitive information has been excluded from this document.

1.19.3. Further Information

No further information needs to be provided at this time.

2.SAFEGUARDS AND STAKEHOLDER ENGAGEMENT

2.1. Stakeholder Engagement and Consultation

2.1.1. Stakeholder Identification

Stakeholder Identification

Stakeholders identification will take place following FAO's FPIC principles and CCB guidelines.

Blue Coast Carbon will carry out a thorough stakeholder mapping and analysis before the project starts, to identify who has rights and interests over the project area, and how every stakeholder relates to each other and to the project.

We will seek to map both primary and secondary stakeholders. Primary stakeholders include individuals or groups with statutory or customary rights to land or resources within the project area, e.g. Indigenous Peoples and Local Communities, residents in the area, smallholder farmers, and businesses operating in the project area. Secondary stakeholders are local, regional, and national authorities or organizations involved in the region's conservation, restoration, or development initiatives, e.g., Government agencies and traditional or customary power structures, Non-Governmental Organizations (NGOs), Academic or research institutions, non-local businesses operating around the site or in relevant industries, other local carbon project proponents or developers.

Once all the relevant stakeholders are identified, Blue Coast Carbon together with the local partner will analyse living standards, health, access to infrastructures and facilities, climate change adaptation of each group. Each stakeholder will be prioritised and selected based on their importance and influence over the project, to categorise those to be involved, consulted or informed during each project phase. The potential impact of the project on the different groups will also be assessed.

The stakeholders analysis will also include a gender analysis, assessing the relative positions of women and men in terms of representation and influence, and the benefits and disincentives associated with the allocation of tasks to women and men.

Blue Coast Carbon has been working on the ground in Guinea since March 2025. Since then, the Company has carried out a preliminary stakeholder mapping, meeting Government representatives and potential partners. These include the following:

- National government: Ministry of Environment and Sustainable Development (MEDD), Ministry of Agriculture and Livestock, National Parks Directorate (OGPNRF)

	- Public agencies: French Embassy, Service de Coopération et d'Action Culturelle (SCAC), Agence Française de Développement - NGOs: Biotope, United Nation Development Programme, Charente-Maritime Coopération Section to be completed and updated in the PD submitted for validation.
Legal or customary tenure/access rights	Under Articles 53 and 54 of the Maritime Code (2019), all Guinean mangroves fall within the Maritime Public Domain (DPM). This domain is inalienable and includes a 100-meter coastal strip and all areas up to the limit of water salinity. Consequently, while Article 17 of the Forestry Code allows landowners to own forest products, the inalienability of the DPM ensures that mangrove land and its biological resources remain the exclusive property of the State. Despite State ownership, Articles 46–50 of the Forestry Code recognize "usage rights" for local communities. These rights permit subsistence activities, including artisanal fishing, salt production, and domestic wood collection. To bridge the gap between statutory State ownership and customary use, the project will secure tenure through: Temporary occupation permits, legalising project activities within the DPM. Community management agreements, formalising traditional usage rights into sustainable management contracts under Forestry Administration oversight.
Stakeholder diversity and changes over time	Section to be completed in the PD submitted for validation.
Expected changes in well-being	Section to be completed in the PD submitted for validation.
Location of stakeholders	Section to be completed in the PD submitted for validation.

Location of resources

Section to be completed in the PD submitted for validation.

2.1.2. Stakeholder Consultation and Ongoing Communication

Stakeholder consultation has begun at the national level, with multiple meetings throughout 2025 and into 2026 with the Ministry of Environment and Sustainable Development (MEDD). Blue Coast Carbon has also consulted with numerous other stakeholders and potential partners including National Parks Directorate (OGPNRF), Biotope, United Nation Development Programme, Charente-Maritime Coopération.

A local stakeholder consultation will be carried out before the implementation of project activities, to make sure all the stakeholders identified will be involved.

The consultations will be conducted in a manner that is inclusive and respectful of local knowledge.

The full participation of stakeholders in both project design and implementation is key to the project's success. Therefore, the local community will be involved in all project stages, from planning to design and implementation. A stakeholder engagement plan customised to all groups identified, will be co-defined with key stakeholders. It will include at least the following steps:

- introduction of project to all primary stakeholders, discussing project goals and objectives, outlining expected outcomes and the project timeline, and introducing the project team
- gathering feedback from all stakeholders through consultations customised to each target group. At least 2 focus groups will be held separately with only women.
- collection of information necessary for the project development, e.g. traditional uses of mangroves, sacred sites, history of degradation, historical presence of wildlife on the project site, etc.
- obtaining agreement on project design and activities, by means of co-designing activities.

A grievance management procedure will also be designed to deal with any grievance during the engagement process and project implementation.

The project proponent will develop a procedure for monitoring and evaluation of decision-making and engagement processes to allow periodic reviews of the agreement with the community and the engagement process.

All stakeholders engagement activities will be regularly reported throughout the project duration.

Date of stakeholder consultation

Section to be completed in the PD submitted for validation.

Stakeholder engagement process	Section to be completed in the PD submitted for validation.
Consultation outcome	Section to be completed in the PD submitted for validation.
Ongoing communication	Section to be completed in the PD submitted for validation.
Stakeholder input	Section to be completed in the PD submitted for validation.

2.1.3. Free Prior and Informed Consent

Obtaining consent	Blue Coast Carbon is committed to obtain free, prior and informed consent before starting project activities. To achieve this goal, the following steps will be included in the stakeholder engagement plan: • Identify Indigenous Peoples with customary rights and interests over the project area, and engage their representatives, making sure that both men and women are considered • Design a participatory communication plan • Meet with relevant stakeholders and agree on a process for consultation and obtaining consent, respecting traditional and customary protocols and dynamics • Establish a process for dealing with disputes and complaints • arrange meetings with the representative body to introduce team members, explain the proposed carbon project, provide information and negotiate details about the project, confirm the project area, and agree on a process and timeframe for engagement and decision-making. • Gathering consent on the restrictions to activities in the project areas
Outcome of FPIC	Section to be completed in the PD submitted for validation.

Section to be completed and updated in the PD submitted for validation.

2.1.4. Grievance Redress Procedure

A grievance mechanism will be established to address concerns related to the project in the project area. This mechanism will align with international best practices and will include steps to identify potential risks, concerns, and conflicts associated with the project. The system will be designed to enable Indigenous Peoples and local communities to express their concerns and grievances effectively. An appropriate grievance mechanism will be drafted to respond to grievances and integrate feedback during the project's design and implementation phases. A grievance register will be maintained to document all complaints and how they have been addressed and solved. The information collected will be used to introduce changes in the project to reduce dispute-related risks and to learn for future projects, and to improve FPIC processes as the project progresses.

Section to be completed and updated in the PD submitted for validation.

2.1.5. Public Comments

Section to be completed in the PD submitted for validation.

2.2. Risks to Stakeholders and the Environment

2.2.1. Management Experience

Blue Coast Carbon was founded to develop the most reliable nature-based projects for high-integrity carbon credits. The company's mission is exploiting carbon markets to support the sustainable development of low-income Countries. The company specialises in blue carbon projects to contribute to the protection of marine environments while mitigating climate change and support community development.

The management team has extensive expertise in geospatial data analysis and remote sensing, marine ecology and cooperative development projects. The team all have over 15 years of experience delivering advanced remote sensing projects using satellite and drone data across more than 40 countries. We leverage proprietary data-driven analysis, coastal ecosystem expertise, and industry-standard verification to originate high-integrity blue carbon credits backed by real-world impact and future resilience. The skill set within Blue Coast Carbon covers finance, project management, environmental science, carbon offset projects, community development, technology, and many other areas, including on-ground experience managing projects and operations in Guinea.

Additional local partners are being identified to support community engagement, rural development, and project activity implementation.

2.2.2. Risk Assessment

Section to be completed in the PD submitted for validation.

2.3. Respect for Human Rights and Equity

2.3.1. Labor and Work

Section to be completed in the PD submitted for validation.

2.3.2. Human Rights

Section to be completed in the PD submitted for validation

2.3.3. Indigenous Peoples and Cultural Heritage

Section to be completed in the PD submitted for validation.

2.3.4. Property Rights

Section to be completed in the PD submitted for validation.

2.3.5. Benefit Sharing

Process used to design the benefit sharing plan	Section to be completed in the PD submitted for validation.
Summary of the benefit sharing plan	Section to be completed in the PD submitted for validation.
Approval and dissemination of benefit sharing plan	Section to be completed in the PD submitted for validation.

2.4. Ecosystem Health

Section to be completed in the PD submitted for validation.

2.4.1. Rare, Threatened, and Endangered Species

Section to be completed in the PD submitted for validation.

2.4.2. Introduction of Species

Only native species of mangroves will be considered for any activity of replanting, based on the natural species composition in the specific site.

Species introduced	Classification	Justification for use	Adverse effects and mitigation
Avicennia germinans	Native	Naturally occurring in the project area - to be completed in a later version of PD	
Laguncularia racemosa	Native	Naturally occurring in the project area - to be completed in a later version of PD	
Conocarpus erectus	Native	Naturally occurring in the project area - to be completed in a later version of PD	
Acrostichum aureum	Native	Naturally occurring in the project area - to be completed in a later version of PD	
Rhizophora harrisonii	Native	Naturally occurring in the project area - to be completed in a later version of PD	

Rhizophora mangle	Native	Naturally occurring in the project area - to be completed in a later version of PD	
Rhizophora racemosa	Native	Naturally occurring in the project area - to be completed in a later version of PD	

Existing invasive species	Mitigation measures to prevent the spread or continued existence of invasive species
to be completed in a later version of the PDD	to be completed in a later version of the PDD

Section to be updated and completed in the PDD submitted for validation.

2.4.3. Ecosystem Conversion

The project area has been extensively analysed by means of historical datasets and satellite images. Such evidence suggests that degradation and clearing of the degraded areas within the project boundaries took place at least 10 years ago.

The section will be completed in the PDD submitted for validation.

3. APPLICATION OF METHODOLOGY

3.1. Title and Reference of Methodology

Type (methodology, tool or module).	Reference ID, if applicable	Title	Version
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Methodology	VM0007	VM0007 REDD+ Methodology Framework (REDD+MF),	1.8
Methodology	VM0033	VM0033 Methodology for Tidal Wetland and Seagrass Restoration	2.1
Module	VMD0055	VMD0055 – Estimation of Emission Reductions from Avoiding Unplanned Deforestation	1.1
Module	VMD0019	Methods to Project Future Conditions	1.0
Tool		AFOLU Non-Permanence Risk Assessment Calculation Tool	4.2
Module	VMD0051	Methods for Monitoring of Carbon Stock Changes and Greenhouse Gas Emissions and Removals in Tidal Wetland Restoration and Conservation Project Activities (M-TW)	1.1
Module	VMD0052	Demonstration of Additionality of Tidal Wetland Restoration and Conservation project activities	2.0
Module	VMD0007	Estimation of baseline carbon stock changes and greenhouse gas emissions from unplanned deforestation and unplanned wetland degradation (BL-UP)	3.3
Tool	CDM AR Tool 1	CDM AR Tool for testing significance of GHG emissions in A/R CDM project activities	1.0
Tool	CDM AR Tool 2	Combined tool to identify the baseline scenario and demonstrate additionality for A/R CDM project activities	1.0

CDM Module	CDM AR Tool 3	Calculation of the number of sample plots for measurements within A/R CDM project activities	2.1
CDM Module	CDM AR Tool 14	Estimation of carbon stocks and change in carbon stocks of trees and shrubs in A/R CDM project activities	4.2

3.2. Applicability of Methodology

Table 4 All REDD+ activity types

Methodology ID	Applicability condition	Justification of compliance
VM0007	Land in the project area has qualified as forest for at least the 10 years prior to the project start date. Mangrove forests are excluded from any tree height requirement in a forest definition, as they consist of 95 % mangrove species, which often do not reach the same height as other tree species. Mangroves occupy contiguous areas and their functioning as a forest is independent of tree height.	The forests in the project area are mangrove forests and are thus excluded from any tree height requirements.
VM0007	Where land within the project area is peatland or tidal wetlands and emissions from the SOC pool are deemed significant, the relevant WRC modules are applied alongside other relevant modules	The relevant WRC modules to account for the SOC pool have been applied.
VM0007	Baseline deforestation in the project area falls within either of the following categories:	The baseline deforestation in the project area falls within the unplanned deforestation (VCS category AUDef) category

	a) Unplanned deforestation (VCS category AUDef) b) Planned deforestation (VCS category APDef)	
VM0007	REDD activity types are not applicable when leakage prevention activities include: a) Flooding agricultural lands to increase production (e.g., rice paddies); and/or b) Intensifying livestock production through use of feed-lots and/or manure lagoons	Leakage prevention activities will not include flooded agricultural lands nor intensifying livestock production through use of feed-lots and/or manure lagoons.

Table 5 Avoiding unplanned deforestation

Methodology ID	Applicability condition	Justification of compliance
VM0007	Baseline agents of deforestation meet all of the following criteria: a) Clear the land for tree harvesting, settlements, crop production (agriculturalist), ranching or aquaculture, where such clearing for crop production, ranching or aquaculture does not amount to large scale industrial agriculture or aquaculture activities; b) Have no documented and uncontested legal right to deforest the land for these purposes; and c) Are either residents in the reference region for deforestation or immigrants.	Baseline agents of deforestation have: a) Cleared the land for tree harvesting for firewood and to convert land for rice and salt production. b) no documented and uncontested legal right to deforest the land for these purposes c) Are residents in the reference region for deforestation

VM0007	Avoiding unplanned deforestation activities are not applicable under the following condition: Post-deforestation land use in the baseline scenario constitutes reforestation.	Post deforestation land uses do not include reforestation
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Table 6 Applicability conditions for WRC activities

Methodology ID	Applicability condition	Justification of compliance
VM0007	WRC activities are not eligible under the following conditions: - Project activities lower the water table, unless the project converts open water to tidal wetlands, or improves the hydrological connection to impounded waters. - Changes in hydrology do not result in the accumulation or maintenance of SOC stock, noting that a) this pertains to projects that intend to sequester carbon through sedimentation and/or vegetation development and b) this does not pertain to projects that increase salinity to reduce CH₄ emissions. Projects that aim to decrease CH₄ emissions through increased salinity must account for any changes in SOC stocks. - Hydrological connectivity of the project area with adjacent areas leads to a significant increase in GHG emissions outside the project area.	Project activities will not lower the water table in the project conservation areas. The hydrological restoration will be designed so that zones outside the project restoration areas are not affected. The changes in hydrology that could happen will not result in accumulation of SOC by themselves. Project activities do not include the burning of soil. Nitrogen fertilizers will not be used.

	• Project activities include the burning of organic soil. • Nitrogen fertilizer(s), such as chemical fertilizer or manure, are applied in the project area during the project crediting period.	
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Table 7 Applicability conditions for RWE activities

Methodology ID	Applicability condition	Justification of compliance
VM0007	Prior to the project start date, the project area meets one of the following criteria: a) The project area is free of any land use that may be displaced outside the project area, as demonstrated by at least one of the following, where relevant: i) The project area has been abandoned for two or more years prior to the project start date; or ii) Use of the project area for commercial purposes (i.e., trade) is not profitable as a result of salinity intrusion, market forces or other factors. In addition, timber harvesting in the baseline scenario within the project area does not occur; or iii) Degradation of additional wetlands for new agricultural/aquacultural sites within the country will not occur or is prohibited by enforced law.	Section to be completed in the PD submitted for validation.

	OR b)The area is under a land use that may be displaced outside the project area, although in such case, baseline emissions from this land use must not be accounted for, and where degradation of additional wetlands for new agricultural/aquacultural sites within the country will not occur or is prohibited by enforced law. OR c)The area is under land use that will continue at a similar or greater level of service or production during the project crediting period (e.g., reed or hay harvesting, collection of fuelwood, subsistence harvesting, commercial fishing). The project proponent must demonstrate (a), (b) or (c) above, based on verifiable information such as laws and bylaws, management plans, annual reports, annual accounts, market studies, government studies or land use planning reports and documents.	
VM0007	Project activities restoring tidal wetlands include any of the following, or a combination of the following: a) Creating, restoring and/or managing hydrological conditions (e.g., removing	The restoration project activities include: - Restoration of hydrological conditions to allow for natural vegetation re-colonisation - Re-introducing native plant communities, where natural re-colonisation is not possible.

	tidal barriers, improving hydrological connectivity, restoring tidal flow to wetlands or lowering water levels on impounded wetlands) b) Altering sediment supply (e.g., beneficial use of dredge material or diverting river sediments to sediment-starved areas) c) Changing salinity characteristics (e.g., restoring tidal flow to tidally restricted areas) d) Improving water quality (e.g., reducing nutrient loads leading to improved water clarity to expand seagrass meadows, recovering tidal and other hydrologic flushing and exchange or reducing nutrient residence time) e) (Re-)introducing native plant communities (e.g., reseeding or replanting) f) Improving management practice(s) (e.g., removing invasive species, reduced grazing) g) Prescribed burning of herbaceous and shrub aboveground biomass (cover burns)	This section will be completed in the PD submitted for validation.
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Table 8 Applicability conditions for CIW activities

Methodology ID	Applicability condition	Justification of compliance
VM0007	Project activities conserving tidal wetlands include: a) Protecting at-risk wetlands (e.g., establishing conservation easements, establishing community supported management agreements, establishing protective government regulations, and preventing disruption of water and/or sediment supply to wetland areas) b) Improving water management on drained wetlands c) Maintaining or improving water quality for seagrass meadows d) Recharging sediment to avoid drowning of coastal wetlands e) Creating accommodation space for wetlands migrating with sea-level rise	Project activities aimed at conserving mangrove forests on tidal wetlands. All the necessary activities to protect at-risk mangrove forests will be undertaken. This section will be completed in the PD submitted for validation.
VM0007	Baseline agents of wetland degradation in avoiding unplanned wetland degradation activities meet all of the following criteria: ● Cause an alteration in the hydrology of the project area (involving drainage, interrupted sediment supply or both) and/or a loss of soil organic carbon ● Have no documented and uncontested legal right to degrade the wetland, and ● Are either residents in the reference region for wetland degradation or immigrants.	Section to be completed in the PD submitted for validation.

Table 9 Applicability conditions for tidal wetland restoration activities

Methodology ID	Applicability condition	Justification of compliance
VM0033	Project activities which restore tidal wetlands are eligible	Project activities aim at restoring degraded tidal wetlands with mangrove forests. The project area qualifies as a tidal wetland as defined in VM0033. It is subject to regular tidal inundation and characterised by saline and brackish water conditions.

VM0033	Project activities may include any of the following, or combinations of the following: a) Creating, restoring and/or managing hydrological conditions (e.g., removing tidal barriers, improving hydrological connectivity, restoring tidal flow to wetlands or lowering water levels on impounded wetlands) b) Altering sediment supply (e.g., beneficial use of dredge material or diverting river sediments to sediment starved areas) c) Changing salinity characteristics (e.g., restoring tidal flow to tidally restricted areas) d) Improving water quality (e.g., reducing nutrient loads leading to improved water clarity to expand seagrass meadows, recovering tidal and other hydrologic flushing and exchange, or reducing nutrient residence time) e) (Re-)introducing native plant communities (e.g., reseeding or replanting) f) Improving management practice(s) (e.g., removing invasive species, reduced grazing) Prior to the project start date, the project area: a) Is free of any land use that could be displaced outside the project area, as demonstrated by at least one of the following, where relevant: - i)The project area has been abandoned for two or more years prior to the project start date; or	Project activities will restore hydrological conditions and tidal flow to help the natural recolonisation of the area by endemic mangrove species. Replanting will be carried out only where natural recolonisation is not possible. Prior to the start date, the project area is free of any land use as its use is not profitable as it includes remote areas and land that is difficult to farm.
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ii) Use of the project area for commercial purposes (i.e., trade) is not profitable as a result of salinity intrusion, market forces or other factors. In addition, timber harvesting in the baseline scenario within the project area does not occur; or

iii) Degradation of additional wetlands for new agricultural sites within the country will not occur or is prohibited by enforced law.

OR

b) Is under a land use that could be displaced outside the project area), although in such case baseline emissions from this land use must not be accounted for, and where degradation of additional wetlands for new agricultural/aquacultural sites within the country will not occur or is prohibited by enforced law.

OR

c) Is under a land use that will continue at a similar level of service or production during the project crediting period (e.g., reed or hay harvesting, collection of fuelwood, subsistence harvesting).

VM0033	Live tree vegetation may be present in the project area and may be subject to carbon stock changes (e.g., due to harvesting) in both the baseline and project scenarios.	Mangrove vegetation is present in the project area and is subject to carbon stock changes. Project activities are aimed at reducing mangrove harvesting by providing alternative income to local communities.
VM0033	The prescribed burning of herbaceous and shrub aboveground biomass (cover burns) as a project activity may occur.	The project activities do not include the prescribed burning of biomass
VM0033	Where the project proponent intends to claim emission reductions from reduced frequency of peat fires, project activities must include a combination of rewetting and fire management.	The project area does not include peatlands
VM0033	Where the project proponent intends to claim emission reductions from reduced frequency of peat fires, it must be demonstrated that a threat of frequent on site fires exists, and the overwhelming cause of ignition of the organic soil is anthropogenic (e.g., drainage of the peat, arson).	The project area does not include peatlands
VM0033	In strata with organic soil, afforestation, reforestation, and revegetation (ARR) activities must be combined with rewetting.	The project area is not featured by organic soil
VM0033	Project activities do not qualify as IFM or REDD.	Project activities do not qualify as IFM, as the project area is not a managed forest, and no harvesting will happen.

		REDD project activities take place in combination with WRC and ARR, therefore both VM0033 and VM0007 are applicable.
VM0033	Baseline activities do not include commercial forestry.	No commercial forestry is undertaken in the project area nor in the baseline area.
VM0033	Project activities do not lower the water table, unless the project converts open water to tidal wetlands, or improves the hydrological connection to impounded waters.	Project activities will not lower the water table
VM0033	Hydrological connectivity of the project area with adjacent areas does not lead to a significant increase in GHG emissions outside the project area.	The project activities do not cause any alteration of the hydrology of the project area.
VM0033	Project activities do not include the burning of organic soil.	Project activities do not include the burning of organic soil.
VM0033	Nitrogen fertilizer(s), such as chemical fertilizer or manure, are not applied in the project area during the project crediting period.	The project activities do not foresee the use of any fertiliser

3.3. Project Boundary

Source		Gas	Included?	Justification/Explanation
Baseline	Above-ground tree biomass	CO ₂	Yes	AGB in the baseline scenario is considered only in tree-covered strata and is not expected to change due to forest maturity.

Source		Gas	Included?	Justification/Explanation
	Above-ground non-tree biomass	CO ₂	No	The project's baseline scenario is characterized by mangrove forests and degraded wet lands, where the aboveground non-tree biomass carbon pool is not present in a significant or quantifiable manner. Furthermore, following the implementation of restoration activities, this pool is projected to continue making a negligible contribution to the overall ecosystem carbon stocks.
	Below-ground biomass	CO ₂	Yes	Belowground biomass in the baseline scenario is considered in the tree-covered strata, and it is not expected to change.
	Deadwood and litter	CO ₂	No	This pool is optional
	Soil organic carbon	CO ₂	Yes	SOC in the baseline scenario is considered and it is not expected to change.
	Burning of woody biomass	CO ₂	No	No biomass burning is anticipated to take place in the baseline scenario. The project does not include the burning of biomass for soil preparation.
Project	Above-ground tree biomass	CO ₂	Yes	ABG in the project scenario is expected to increase
	Above-ground non-tree biomass	CO ₂	No	To maintain a conservative approach and to align with the baseline accounting, the aboveground non-tree biomass pool has been excluded from the quantification of emission reductions for this project.

Source	Gas	Included?	Justification/Explanation
Deadwood and litter	CO ₂	No	This pool is optional
Soil organic carbon	CO ₂	Yes	SOC in the project is considered and it is expected to change. The implementation of project activities is expected to enhance soil organic carbon stocks.
Burning of woody biomass	CO ₂	No	No biomass burning is anticipated to take place in the project scenario. The project does not include the burning of biomass for soil preparation.

The project boundaries of the first project instance include a total of 73,751 ha of stable mangrove forests. The location of conservation and restoration project areas currently under consideration are presented in Figure 7, shaded in green and orange respectively.

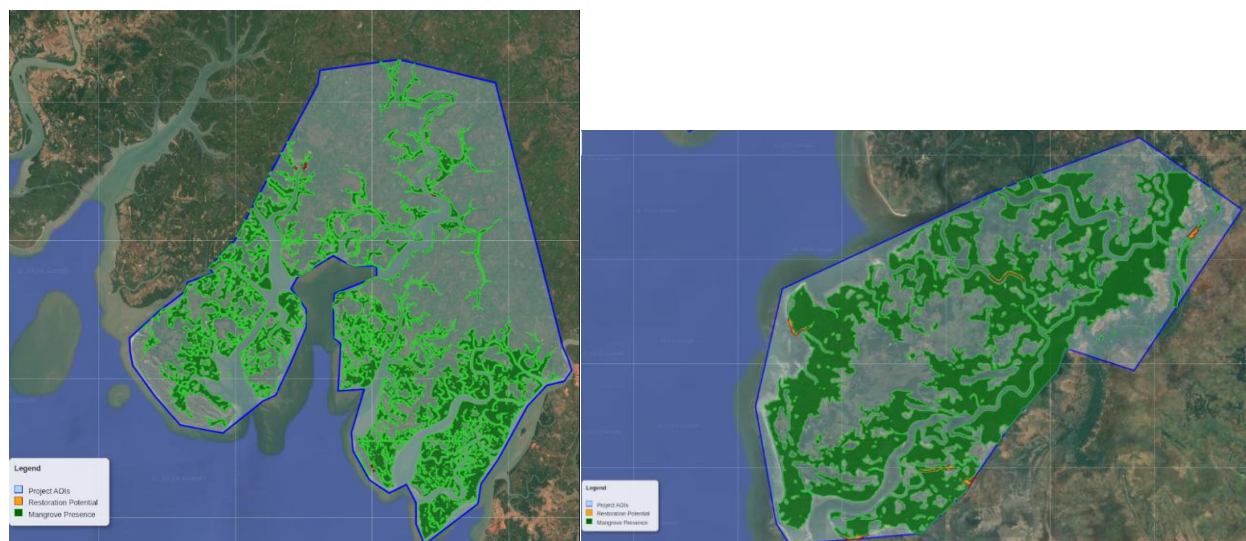


Figure 7 Project boundaries of the first instance showing restoration (orange) and conservation (green) areas.

3.4. Baseline Scenario

Section to be completed in the PDD submitted for validation.

3.5. Additionality

Tidal wetland restoration and conservation projects shall apply module VMD0052 'Demonstration of additionality of tidal wetland restoration and conservation project activities'. According to the module, a project is additional if it meets the requirements for regulatory surplus, and the applicability conditions (positive list).

3.5.1. Regulatory Surplus

Is the project located in an UNFCCC Annex 1 or Non-Annex 1 country?

- ☐ Annex 1 country ☒ Non-Annex 1 country

Are the project activities mandated by any law, statute, or other regulatory framework?

- ☐ Yes ☒ No

If the project is located inside a Non-Annex 1 country and the project activities are mandated by a law, statute, or other regulatory framework, are such laws, statutes, or regulatory frameworks systematically enforced?

- ☐ Yes ☒ No

There are no specific legislations for mangrove forests in Guinea. Nonetheless, mangrove's importance for climate, biodiversity and society is recognized by state actors and policy frameworks are being designed both at national and regional levels for this habitat.

The Ministry of Environment, Water and Forestry is in charge of:

- The Master Plan for the Development of Mangroves;
- consultations with communities in mangrove areas affected by a high rate of deforestation;
- Proposing solutions for mangrove degradation.

However, Guinea has not established dedicated coastal protected areas yet, and mangrove forests are only partially included in the existing protected areas. This is mostly due to the following:

- Lack of capacity of governmental institutions, limited training, lack of enforcement, lack of resources.
- Laws governing protected areas are often ambiguous. For example, the Ministry of Environment is legally responsible for managing classified forests, yet in practice the préfet is also allowed to grant extraction rights to individuals, although not for mangrove extraction.
- Protected areas are poorly defined; they include classified forests, national parks, etc. but not everyone agrees with this definition.

The proposed project activities are not specifically mandated by any regulatory framework and the more generalist conservation frameworks are not currently being enforced in the coastal areas, which demonstrates the regulatory surplus of the project activities.

3.5.2. Additionality Methods

The additionality procedure as per module VMD0052 consists of two steps. Step 1 consists in demonstrating regulatory surplus (see Section 3.5.1), and Step 2 requires the project to meet the applicability conditions of the module.

The proposed project is compliant with such conditions as project activities restoring tidal wetlands include the following:

- restoring hydrological conditions (e.g., removing tidal barriers, improving hydrological connectivity, restoring tidal flow to wetlands) to assist natural recovery of vegetation;
- re-introducing native mangrove communities by means of replanting, where natural recovery is not possible;
- improving management practices by means of removing invasive species.

Furthermore, project activities conserving tidal wetlands include protecting at-risk mangrove forests from degradation and deforestation by means of conservation programmes, community awareness campaigns, and promotion of alternative livelihoods and sustainable wood uses to reduce pressure on mangroves.

Therefore, the proposed project can be considered additional.

3.6. Methodology Deviations

Section to be completed in the PD submitted for validation.

4. QUANTIFICATION OF ESTIMATED GHG EMISSION REDUCTIONS AND REMOVALS

4.1. Baseline Emissions

Section to be completed in the PD submitted for validation.

4.2. Project Emissions

Section to be completed in the PD submitted for validation.

4.3. Leakage Emissions

Section to be completed in the PD submitted for validation.

4.4. Estimated GHG Emission Reductions and Carbon Dioxide Removals

Section to be completed in the PD submitted for validation.

5. MONITORING

5.1. Data and Parameters Available at Validation

Section to be completed in the PD submitted for validation.

5.2. Data and Parameters Monitored

Section to be completed in the PD submitted for validation.

5.3. Monitoring Plan

Section to be completed in the PD submitted for validation.